



Department of Telecommunications
Telecommunication Engineering Centre
K.L. Bhawan, Janpath, New Delhi – 110001
(Convergence & Broadcasting Division)



No: - 4-16/2018-C&B/TEC/NWG-9`

dated: 13.10.2022

Subject: Minutes of the fifth meeting of the National Working Group 9 (NWG-9), corresponding to ITU-T Study Group 9 (SG9) for the Study Period 2022- 2024, held on 11-10-2022.

The fifth meeting of the National Working Group-9 (NWG-9) corresponding to the ITU- T Study Group 9 (SG9) reconstituted for the Study Period 2022-24 regarding "Audio visual content transmission and integrated broadband cable networks" or "Broadband Cable and TV" was held on 11-10-2022 under the chairmanship of Sh. Avinash Agarwal, DDG (Convergence & Broadcasting), TEC through video- conferencing. List of participants is placed at Annexure-I.

2. At the outset, Sh. Rakesh Desai, Director (C&B-II) TEC & Convener of NWG-9, welcomed all the participants and briefly explained about the discussions held during the previous four meetings. He specially welcomed Sh. Neeraj Kumar, DDG (National Broadband Mission), DOT & Coordinator of NWG-9 and other new participants from the C-DAC Pune, BIF, MIB, academia, BECIL, etc and explained them briefly about the activities of NWG-9. He informed the members about the Agenda of the meeting and outcome of the first ITU-T SG-9 meeting held virtually from 6-14th September, 2022 including approved work-items in respect of NWG-9 Contribution.

3. Shri Avinash Agarwal, Chairman NWG-9 & DDG (C&B) TEC, in his introductory remarks, stated that contributions of the NWG-9 members in respect of Q.1/9, 2/9 and 11/9 were agreed upon by SG9 in its meeting held on 6-14th September and these contributions resulted into initiating four new work items – one Recommendation on CAS, and three Technical Reports, one each on CAS, Wi-Fi TV, and Live Captioning. He informed that NWG-9 Members have been assigned various management positions by ITU-T SG-9, as mentioned below:

- a. **Prof Dr. Pradipta Biswas, Indian Institute of Science, India**
 - SG9 Vice-chairman; Vice-chairman of WP2 of SG9, Co-chairman of IRG-AVA; Liaison Officer for JCA-AHF.
- b. **Sh. Avinash Agarwal, TEC:**
 - Rapporteur Q11/9; Associate Rapporteur Q3/9; Liaison Officer from SG9 to SG3; Editor of 4 work items – J.FSR-REQ, TR.FSR, J.TR.WiFiTV, JSTR.LCAP.
- c. **Sh. Rakesh Desai, TEC:**
 - Rapporteur Q8/9; Editor of 2 work items – J.FSR-REQ, TR.FSR
- d. **Ms. Meenakshi Singhvi, Prasar Bharati:**
 - Associate Rapporteur Q.6/9
- e. **Sh. Rajiv Khattar**
 - Associate Rapporteur Q2/9; Editor of 1 work item – JSTR.LCAP
- f. **Sh. Pallab Dutta, CDOT:**
 - Editor of 1 work item – J.TR.WiFiTV

4. Shri Avinash Agarwal also appreciated the work done by the NWG-9 members in respect of ITU-T SG9 contributions and highlighted that the management team of SG9 has specifically praised the members of the Indian delegation for successfully discharging their responsibilities in respect of the aforesaid roles and enthusiastic and active participation during SG9 meeting. He further mentioned that such appreciation to the Indian delegation by ITU-T SG9 could not be possible without the collaborative effort of the entire NWG-9.

5. After the opening remarks, on request of the Chair, all the participants formally introduced themselves. Thereafter, open house discussions were initiated on the following new work items agreed upon by SG-9:

- i. J.TR.WiFiTV (Technical Report) on Secondary distribution of digital television and audiovisual content to portable devices using Wi-Fi (Q.1/9);
- ii. J.FSR REQ (Draft Recommendation) Factual subscriber base reporting and protected content delivery in conditional access system (CAS) -Requirements (Q.2/9);
- iii. TR-FSR (Technical report) Factual subscriber base reporting and protected content delivery in conditional access system (Q.2/9);
- iv. JSTR.LCAP (Technical Report) Technical advances, challenges, and best practices in live captioning (Q.11/9).

6. Sh. Avinash Agarwal explained the group on the above-stated work-items in respect of Q.2/9 & 11/9. Sh. Pallab Dutta explained the work item in respect of Q.1/9.

7. Ms. Meenakshi Singhvi explained the group on NWG-9 contribution submitted during SG9 meeting on the draft recommendation vide 'J.BD2M' regarding "Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks" and the observations / concerns of SG9 thereof. She pointed out that SG-9 has suggested to enhance the said contribution considering the scope of SG-9 so that it may help reaching consensus at Q6/9 level and such updated contribution will be discussed in the future meeting of SG9.

8. After discussions and deliberations, it was decided to form the following three (3) Working Teams for each work item based on the nominations received from the participants:

a) J.TR.WiFiTV (Q.1/9):

- i. Sh Pallab Dutta, C-DOT
- ii. Sh. Manavdeep Singh, C-DOT
- iii. Sh. Arun Mukarji, BIF
- iv. Sh. Aftab Alam, ADG (C&B), TEC.

b) TR-FSR (Q.2/9):

- i. Sh. Rajiv Khattar
- ii. Ms. Kirti Gupta, US (Broadcasting Wing), MIB
- iii. Dr. Shiv Kumar, BIF
- iv. (BECIL will revert shortly with nomination of their member, if any)

c) JSTR.LCAP (Q.11/9):

- i. Ms. Lenali Singh, CDAC-Pune
- ii. Sh. Shubhashish Gupta, CDAC-Pune
- iii. Ms. Neema, BIF

- iv. Dr. Ashay Sahasrabudhe
- v. Ms. Meenakshi Singhvi, Director, Prasar Bharti
- vi. Sh. Avinash Agarwal, DDG (C&B) TEC

9. All participants were requested to send their inputs / comments to the above Working Teams **within a fortnight**. The Working Teams were further requested to submit their reports/ drafts to the Chairman NWG-9 within a month **i.e. by 11.11.2022.**

10. Sh. Rakesh Desai, Director, TEC briefly explained the additional new work items (10 nos) submitted by other Member-States of ITU-T during the SG9 meeting that were also agreed upon by ITU-T SG-9 in respect of Q/1/9, 4/9,6/9, 7/9 & 8/9. Members were requested to send their comments / inputs on these work-items of other Member-States by **30th October, 2022** keeping in view the interest of Govt of India in telecommunications, broadcasting, and other ICT areas and in order to present the India's perspective.

11. Sh. Avinash Agarwal also demonstrated the portal that TEC is developing on the lines of ITU-T Portal to facilitate Indian contributions to various global standardisation organisations. While demonstrating the main features of the portal, he stated that this portal will act as a platform to share the information and collaborate on the standards development activities.

12. Sh. Neeraj Kumar, DDG (NBM), DOT emphasized the need to have more representation from academia and industry in NWG-9 and suggested that interventions of the respective ministries may be sought in this regard, if required. Sh. Avinash Agarwal informed that NWG-9 has representation from academia, industry, OEMs, the concerned Ministries, research organisations, and domain experts. He once again requested all the members to identify relevant experts so that such experts could be requested to join the NWG-9 group.

13. The key action points emerged from the meeting are as under: -

- a) Three Working Teams on the new work-items in respect of NWG-9 contributions were formed as discussed at Para 8 above to prepare the draft Technical Reports consolidating the inputs / comments from NWG-9 members.
- b) The Working Teams may submit their drafts by **11th November 2022** for further deliberation and onwards submission to ITU-T within the prescribed time limit after getting approval of the competent authority in D/o Telecommunications.
- c) Members may submit their comments on work items proposed by other countries as discussed at point 10 above by **30th October 2022.**

14. The meeting ended with the vote of thanks to the chair.


(Rakesh Desai)

Convener, NWG-9 & Director, (C&B-II), TEC

Encl.: Annexure-I

To,
All members of the NWG-9 and all the Participants.

Copy for kind information to:

- 1. Sr.DDG & Head TEC, New Delhi

ANNEXURE-I

List of Participants:

1. Sh. Avinash Agarwal, Chairman NWG-9 & DDG (C&B), TEC.
2. Sh. Neeraj Kumar, DDG (National Broadband Mission), DOT & Coordinator of NWG-9
3. Sh. Rakesh Desai, Convener, NWG-9 & Director (C&B-II), TEC.
4. Ms. Meenakshi Singhvi, Prasar Bharti.
5. Ms. Kirti Gupta, US (Broadcasting Wing), MIB
6. Sh. Prateek, BECIL
7. Sh. Rupesh Kumar, BECIL
8. Sh. Pallab Dutta, C-DoT.
9. Sh. Manav C-DoT
10. Ms. Lenali Singh, CDAC-Pune
11. Sh. Shubhashish Gupta, CDAC-Pune
12. Sh. Rajiv Khattar, Researcher & Industry specialist
13. Sh. Debashish Bhattacharya- BIF
14. Sh. Satyaveer Gulati- BIF
15. Dr. Shiv Kumar- BIF
16. Sh. Arun Mukarji- BIF
17. Ms. Nema Sunil Kumar- BIF
18. Sh. Rajith Ali- SIA India
19. Dr. Ashay Sahasrabudhe.
20. Sh. Sunil Gautam, Director (SPPI), DoT.
21. Sh. Brajesh Kumar, Director (C&B-I), TEC.
22. Md. Aftab Alam, ADG (C&B), TEC